

(a) Raw activations

(b) Hadamard: spread energy

(c) Hadamard: spread



(d) PrismQuant: align

PrismQuant

$v_1 \dots v_k$



1_g



align dominant eigendirections
with group-constant directions



----- common offset $\rightarrow$ zero-point

One token $\cdot$ measured group traces

same grouped asymmetric INT4
residual within-group range

(e) Raw: concentrated

(f) Hadamard: spread

(g) PrismQuant: shared offset